

Virtual Private Network

Module 34

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana in this lecture we will study about virtual private networks first of all please see the contents which I shall cover after covering brief introduction about virtual private Network I shall take up its architecture which will be followed by features of openvpn in last relationship between firewall and VPN will be discussed friends today almost all companies have offices and plants scattered over many cities sometimes over multiple countries in the olden days it was common for such companies to lease lines from the telephone company for connectivity between some or all pairs of locations some companies still

do this a network built up from company computers and Leed telep phone lines is known as a private Network an example of private Network connecting three locations is shown in figure private networks work fine and are very secure as no traffic can leak out of company location and Intruders have to physically wir tap the lines to break in which is not so easy to do 2 minutes, 1 second the problem with private networks is that they are very costly when public data networks and later the internet appeared many companies wanted to move their data and voice traffic to the public network but without giving up the security of the private Network this demand soon led to the invention of VPN that is virtual PR private networks

which are overlay networks on top of public networks but with most of the properties of private networks a virtual private Network VPN provides a secure connection between a sender and a receiver over a public known secure network such as the internet they are known as virtual because they are are merely an illusion just as virtual circuits are not real circuits and virtual memory is not real memory a VPN is created by establishing a virtual Point topoint connection through the use of dedicated connections virtual tunnels and traffic encryption on internet as shown in figure a VPN connection across the internet is similar to a van link between websites from a user perspective the extended network resources are accessed

in the same way as resources available within the private Network vpns allow employees to securely access their company's internet while traveling outside the office similarly vpns securely connect geographically separated offices of an organization creating one cohesive Network VPN technology is also used by individual internet users to secure their Wireless transactions to circumvent Geo restrictions and censorship and to connect to proxy servers for the purpose of protecting personal identity and location now let us see types of VPN currently there are three types of vpns while their goal is to use the internet as a private Enterprise backbone Network each of them addresses the needs of a different interest group in the Enterprise the three types of VPN will

be explained now first is Access VPN which provide remote user or mobile user or tele computer or branch office with reliable access to corporate networks second is internet VPN that allow Branch offices to be linked to corporate at headquarters in a secure manner third is Extranet vpns that allow customers suppliers and partners to access corporate internet in a secure manner now we will see VPN architecture a VPN as shown in figure illustrates VPN architecture which consist of four components a VPN client a network access server NES a tunnel terminating device known as VPN server and a VPN protocol in this a client is connected to NES through public switched telephone network pstn network in a typical access VPN

connection a remote user or VPN client initiates a Point to-point protocol PPP connection with the isp's NES via the public switch telephone Network pstn the NES is owned by the ISP and is usually implemented in the isp's point of presence po after the user has been authenticated by the appropriate authentication method the NES directs the packet to the IP tunnel that connects both the NES and the VPN server the VPN server May reside in the isp's po or at the corporate site depending on the VPN model that is implemented the VPN server recovers the packet from the tunnel unwraps it and delivers it to the corporate Network there are two basic categories of VPN tunneling protocols which are used to establish vpns these are Layer

Two and layer three tunneling protocols Layer Two category contains pointto Point tunneling protocol pptp Layer Two forwarding protocol l2f and layer 2 tunneling protocol l2tp whereas in layer three tunneling protocol we have IP security suit IP SEC when both pptp and l2f were submitted to the ietf it was decided to combine the features of both protocols into a common tunneling protocol which is now L2 TP now first we shall see details of layer 2 protocol Layer Two protocols operate at the data link layer or Layer Two of OSI reference model they use the security provided by Point to-point protocol PPP authentication is performed using existing PPP authentication protocols namely password Authentication

Protocol papap and challenge handshake Authentication Protocol chap no specific provision is made for data encryption which may be performed by the user prior to requesting VPN service the journal structure of the tunnel creation process for layer 2 tunneling protocols is shown in figure a client initiates a PPP Connection by dialing up the NES which is typically implemented

in the isp's PO the client then answers and then perform The Logical control protocol LCP negotiation to establish the PPP authenticated connection once the PPP connection is established the NES uses either PPP or chap to authenticate the client using PPP authentication request client in response reply with PPP authentication

reply message if the authentication is successful the NES attempts to open a PPP connection to the VPN server using openvpn request the VPN server will authenticate the NES using p or chap protocol by giving open VPN reply and thus PPP authentication is successful the client and the VPN server will then use the network control protocol NCP to negotiate the network layer protocol this completes the tunneling process and thus VPN is established between client and corporate server so this was how layer 2 protocol work now let us see first type of layer two protocol that is point to point protocol pptp pptp is a protocol developed by Microsoft and a group of network equipment vendors including Ascend communication and 3com

as an extension of PPP it handles only Point to point connections and it does not support point to multi-point Connections in pptp the NES that allows the remote user to initiate a VPN call at ISP is known as the pptp access concentrator PAC and the VPN server at ISP is known as the pptp network server PNS figure illustrates the components of a pptp based VPN pptp does not provide packet by packet encryption instead it relies on PPP's Native encryption capability in pep and chap figure shown illustrates the pptp packet format the PPP payload consists of the data and its original IP and TCP header a pptp packet consisting of PPP payload and PPP header is encapsulated in generic routing encapsulation GR header which is then carried over IP IP header is inserted on

top of GR header now next type of layer two protocol is Layer Two forwarding L2F L2F is a proprietary protocol that was developed by Cisco Systems it is protocol independent and can run over x.25 frame relay and ATM networks it supports private IP IPX and Apple Talk and uses UDP for internet tunneling L2F defines many connections within a tunnel allowing a tunnel to support many connections figure illustrates the components of an L2F based VPN in L2F the VPN server is known as home Gateway rest all components like NAS at ISP IP tunnel and others are same as in journal architecture of any layer 2 protocol L2F uses PPP for dialup user authentication unlike pptp L2F defines its own encapsulation header which is

not dependent on IP and GR this capability permits L2F to work in different types of network figure illustrates the format of an L2F packet the slip or PPP payload is encapsulated in an L2F packet with an L2F header and an optional L2F checksum as the trailer next type of layer two protocol which we will take up is Layer Two tunneling protocol L2TP as stated earlier L2TP combines the features of pptp and L2F unlike pptp which runs over TCP L2TP runs over UDP and does not use GR because many firewalls do not support GR L2TP is more firewall friendly than pptp figure illustrates the components of an L2TP based VPN in L2TP the NAS is known as the L2TP access concentrator LA and the VPN server is known as the

l2tp network server LNS rest all components are same as in journal architecture of any layer 2 protocol l2tp uses PPP dialup links and as a result also uses PPP and chap for authentication however it allows the use of radius for user authentication l2tp permits multiple tunnels to be created between the same pair of end points allowing the user to create different tunnels with different qualities of service between the same end points figure shows the format of an l2tp packet as l2tp support UDP so there is UDP header after PPP header after UDP there is a IP header l2tp is supported 17 minutes, 1 second by many vendors and is expected to be the pre-dominant protocol once it becomes a standard l2tp relies on IPC to perform data

encryption if l2tp discovers that IPC is not supported at the remote end it uses the less secure PPP encryption after discussing Layer Two protocols let us move on to layer three tunneling protocols that is IP SEC IP SEC was originally designed to add security to the TCP IP IPC provides packet level authentication integrity and confidentiality by adding two security headers the authentication header ah which provides header integrity and authentication without confidentiality and the encapsulating security payload ESP which provides Integrity authentication and confidentiality to payload ESP permits packet by packet encryption and uses a standard-based encryption key management protocol thus IPC VPN can be created

with either ah or ES or both ah does not provide data encryption and is useful in those environments where only authentication is required more importantly since authentication is not regulated ah is the preferred method for vpns ah also has a lower processing overhead than ESP however when data encryption is desired ESP is used on comparison of layer 2 and layer three tunneling protocol it is found that ipx supports only IP however pptp l2f and l2tp can support known IP traffic such as ipx and apple talk unlike IPC Layer Two tunneling protocols support individual dialup assess because they use PPP user authentication which permits seamless dialup connections through isps IP SE is designed for security

20 minutes, 1 second protection between routers and firewalls but it does not provides user authentication so these were Layer Two and layer three tunneling protocols there are few other VPN products that neither supports IPC nor l2tp one such product is openvpn this which is a fairly new opsource technology that uses the open SSL library and SSL version 3 or TLS version 1 protocols along with an mlgm of other Technologies to provide a strong and reliable VPN solution openvpn neither supports layer 2 nor layer three tunneling protocols openvpn support SSL and as such is not compatible with IPC l2tp or pptp one of its major strength is that it is highly configurable and although it runs best on a UDP Port it can be set

to run on any port including TCP Port 440 three another advantage of openvpn is that the open SSL Library used to provide encryption supports a number of cryptographic algorithms like AES Blowfish 3des cast 128 chamelia and many more with openvpn we can tunnel any IP sub Network or virtual ethernet adapter over a single 22 minutes, 1 second UDP or TCP Port configure a scalable load balanced VPN server using one or more machines which can handle thousands of dynamic connections from incoming VPN clients use all of the encryption

authentication and certification features of the open SSL library to protect any private Network traffic on internet use any Cipher key size or hmac digest for datagram integrity checking

supported by the open SSL Library choose between static key based conventional encryption or certificate based public key encryption use static pre-shared keys or tls-based dynamic key exchange use realtime adaptive link compression and traffic shaping to manage link bandwidth utilization tunnel networks whose public end points are Dynamic such as DHCP or dialing clients tunnel networks through connection oriented State full firewalls without having to use explicit firewall rules tunnel networks over net create secure ethernet Bridges using virtual tap devices and control openvpn using a GUI on Windows or Mac operating system compared to pptp and l2tp or IP secc openvpn can be bit fidly to setup when

using generic openvpn software in particular it is necessary to not only download and install the client but also to download and set up additional configuration files many VPN providers get around this configuration Problem by supplying customized VPN clients so friends we have studied many types of VPN in this lecture now let us Summarize each one by one pptp is very insecure and should therefore be avoided while its ease of setup and attractive crossplatform compatibility l2tp or IP SEC has the same advantage and is much more secure l2tp or IP SEC is a good VPN solution for non critical use however for a quick VPN setup without the need to install extra software it remains useful particularly for mobile devices where

openvpn support remains somewhat patchy openvpn is easily the best allround VPN solution despite needing thirdparty software on all platforms it is reliable fast and most importantly secure although it usually needs a bit more settings than the other protocols so basically wherever possible we should always choose open VPN if we need a quick and simple solution such as for protecting our phone from casual criminals or when connecting to public Wi-Fi hotspot then l2tp or IPC will probably do but given the increase using availability of openvpn apps for mobile devices especially Android we still should prefer to use openvpn so far in this lecture we have been studying about VPN actually firewalls and VPN go hand in hand vpns

open secure Tunnels for communication whereas firewalls sit at strategic locations on the network and block certain type of traffic many firewall products provide encrypted firewall to firewall tunnels firewall control assess to corporate network resources and establish trust between the user and the network consider the network configuration in figure shown the firewall at each Network controls access to resources in the network however the data transmitted between the two sides is still vulnerable to attack as it traverses the internet for this purpose vpns are created to provide privacy between two sites as there is usually no trust between the two sides in this way a combination of fireballs and a VPN

28 minutes, 1 second establishes trust and provides privacy between the two sides this approach provides more security than using either firewalls at both sides or a VPN between the two sides figure shows a VPN tunnel between two firewalls in the past firewall products provided only firewall Security Service however many new firewall products now support VPN functionality as stated earlier both firewall functionality and VPN functionality are needed to establish effective security control so friends this was about virtual private Network in this lecture we studied introduction about VPN which was followed by two architectures of VPN that is layer 2 and layer three then we discussed other VPN product which was based on

SSL in the end relationship between firewall and VPN was discussed thank you